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EXPERIMENT 12

Exp No:12 Experiment to detect outliers in a given data set.

AIM:
To detect outliers in a given data set.

DESCRIPTION:
Understand the procedure to identify the outliers in a given dataset

CODE:

```
import numpy as np
array = np.array([
    10, 12, 15, 18, 20, 22, 24, 25,
    27, 30, 32, 35, 38, 40, 42, 100
])
print(array)
print(array.mean())
print(np.percentile(array, 25))
print(np.percentile(array, 50))
print(np.percentile(array, 75))
print(np.percentile(array, 100))
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)

print(lr, ur)
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
sns.displot(array)
plt.show()
new_array = array[(array > lr) & (array < ur)]
print(new_array)
sns.displot(new_array)
plt.show()
lr1, ur1 = outDetection(new_array)
print(lr1, ur1)
final_array = new_array[(new_array > lr1) & (new_array < ur1)]
print(final_array)
```

OUTPUT:

[10 12 15 18 20 22 24 25 27 30 32 35 38 40 42 100]

30.625

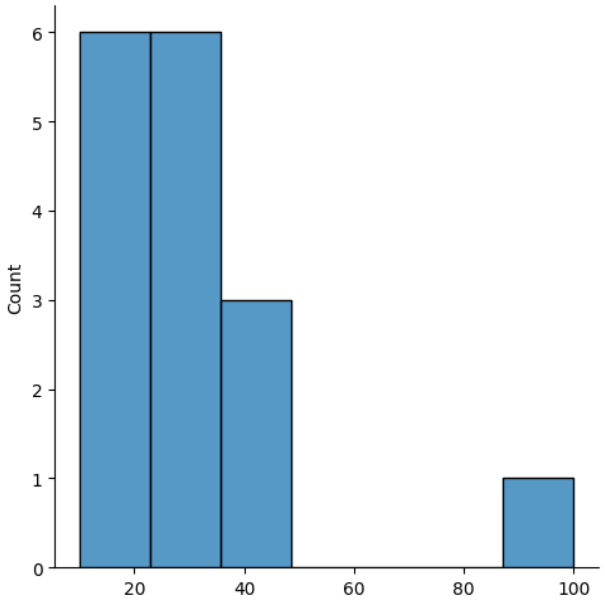
19.5

26.0

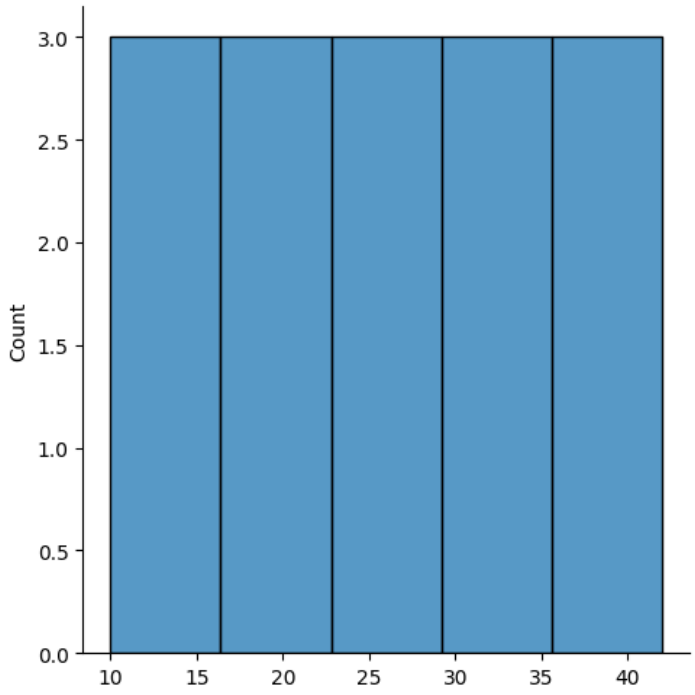
35.75

100.0

-4.875 60.125



[10 12 15 18 20 22 24 25 27 30 32 35 38 40 42]



-2.75 55.25

[10 12 15 18 20 22 24 25 27 30 32 35 38 40 42]

RESULT:

Thus, outliers in the given dataset were successfully detected using the Interquartile Range (IQR) method. The outliers were removed, and the filtered dataset was visualized using histograms.